

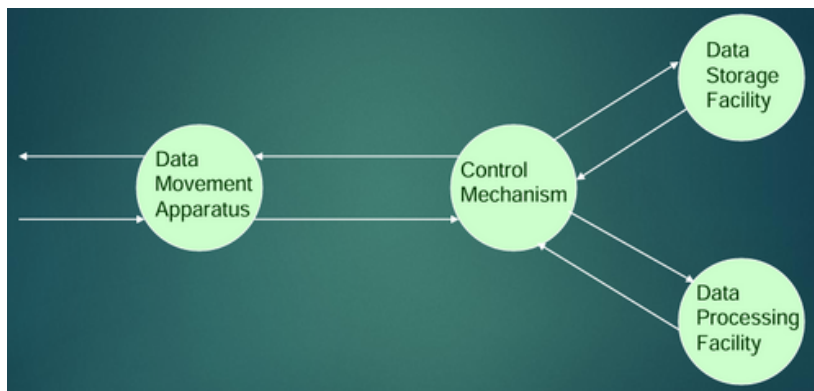
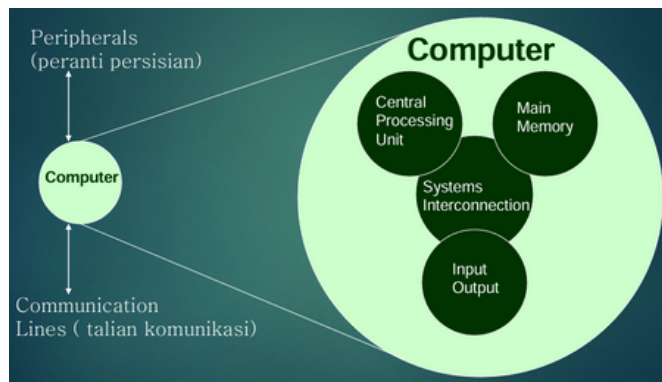
COMPUTER SYSTEM

COMPUTER?

- “Computer” was taken from Latin word *Computare* meaning “calculate”.
- A machine that executes instructions given by user and the data will be processed to produce information.

STRUCTURE AND FUNCTION

- Structure: how components communicate
- Function: the operation performed



Data movement: Data Movement Apparatus => Control Mechanism => Data Movement Apparatus

Data storage: Data Movement Apparatus => Control Mechanism => Data Storage Facility => Control Mechanism => Data Movement Apparatus

Data processing: Data Storage Facility => Control Mechanism => Data Processing Facility => Data Storage Facility

Data processing: Data Movement Apparatus => Control Mechanism => Data Processing Facility => Data Storage Facility and reverse

- **DATA:** Raw fact (Text, Graphic, Audio, Video, Animation)
- **INFORMATION:** Processed data that contains meaning.

ARCHITECTURE

- It refers to the parts of a computer that programmers can see and use, such as the instruction set, data format, addressing, and input/output.

ORGANIZATION

- It refers to how computer hardware components are connected, integrated, and communicate with each other.

COMPUTER CLASSIFICATION

- Microcomputer system
- Minicomputer system
- Mainframe system
- Supercomputer system

COMPUTER GENERATION

- Vacuum tube : 1946 - 1957
- Transistor: 1958 - 1963
- Small scale: 1963 - 1970
- Medium scale: 1971
- Large scale: 1971 - 1977
- Very large scale: 1978 - now

TWO PILLARS OF COMPUTING

1. Universal Computing Devices:

All computers – whether a smartphone, laptop, or supercomputer – are capable of performing the same types of calculations if they are given enough time and memory.

2. Problem Transformation:

Computing is about transforming a real-world problem into something a computer can understand and process – converting ideas into electrical signals.